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Research Article

Contribution of Arab world in transplant research: A PubMed-based bibliometric analysis

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ABSTRACT

Background: The Arab world consists of 22 countries situated in the Middle East and North Africa, tied together by linguistic and cultural bonds. Over numerous past decades, this region tended to fall behind regarding biomedical research. Organ, tissue, and cell transplantation are life-saving and life-improving treatments for various diseases. Given the positive correlation between research activity and the improvement of the clinical frame of transplantation, this article intended to examine the regional activity of transplantation research between 2005 and 2019.

Methods: Using the PubMed database, the number of transplantation-related articles published by each country, and regarding 26 different procedures, was assessed. The results were normalized with respect to each country's average population & average Gross Domestic Product (GDP). In addition, the co-occurrence of keywords and the co-authorships were analyzed by VOS Viewer.

Results: Arab countries contributed to 1.25% of total transplantation publications. The number of transplantation publications had an inflection in 2013, with the last six years alone contributing to 65.27% of the overall regional transplantation-related publications. Kuwait and Lebanon ranked first in terms of publications per million persons, while Egypt ranked first in terms of publications per national GDP. Stem cell, bone & kidney transplantations had the highest number of Arab transplantation-related publications. Low levels of collaboration between authors and organizations were observed, besides a modest but increasing trend towards experimental work on animals and newer therapies.

Conclusion: Despite the increase in transplant-related research activity in recent years, the Arab world still lags behind in this field compared to the worldwide contribution.

1. Introduction

Human organ and tissue transplantation is a very complex medical procedure. This practice requires numerous specialists who join forces to reach the clear-cut diagnosis, understand the essential health issues that make this surgery a necessity, and deliver the best outcomes during the procedure and the post-transplantation recovery phase [1].

For patients who suffer from end-stage organ failure, especially heart and liver failure, organ transplantation from a deceased body may often be the only life-saving treatment [2]. Besides, transplantation can be the cure for end-stage renal diseases requiring kidney transplantation, which is the most common solid organ transplantation procedure, with grafts taken either from brain-dead donors or from live donor transplants, thanks to the advances in this field [3].

Similarly, cell and tissue donation can play a crucial role in restoring essential functions and decreasing disease-related mortality [4]. For instance, umbilical cord blood transplants are currently used in treating various immune and blood-related diseases and genetic disorders [5], which are very common in the Arab world [6].

Although transplantation has seen worldwide progress and success,

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there is a significant fluctuation between countries and regions with regard to the access, safety, and efficacy of these procedures [7]. Availability and differences in transplantation procedures can be observed in the Middle East and are highly affected by the prevailing socio-economic, health, and religious factors [8]. Specifically, we mention the crucial role of religion in government legislation and personal life of citizens in the Arab world [9] and its influence on the law and ethics of Islamic societies [10], which may negatively impact the populations' attitudes towards these procedures [11], leading to a higher gap between donors and recipients [12].

To better understand the regional barriers in the transplantation field in the Arab world that could result in donor-recipient mismatch and graft rejection, this study aims to assess the research activity of Arab countries and their contribution to transplant literature during a 15-year period and to outline the discrepancies in research in the aim of refining clinical practice and improving access to transplant procedures in the Middle East [13].

2. Methods

2.1. Database and search strategy

Screening for transplant-related publications during a period of 15 years (2005 till 2019) was done for all 22 Arab countries, constituted of Algeria, Bahrain, Comoros, Djibouti, Egypt, Iraq, Jordan, Kuwait, Lebanon, Libya, Mauritania, Morocco, Oman, Palestine, Qatar, Saudi Arabia, Somalia, Sudan, Syria, Tunisia, United Arab Emirates, and Yemen.

Using Boolean operator (AND, OR, NOT), we searched PubMed for "transplant" Mesh Term, "name of Arab country required" affiliation, and "2005 - 2019" Mesh Date of publication. Cities in the United States called Lebanon were excluded, and regarding Palestine, we used West Bank and Gaza affiliations.

2.2. Inclusion and exclusion criteria

This study included publications from the PubMed database, conducted between 2005 and 2019, with at least an author affiliated to an Arab country.

2.3. Interpretation and comparison

The average population and GDP were obtained for each country between 2005 and 2019 using the 2019 World Prospect Population and the 2019 World Bank, respectively [14,15]. For each country, the number of publications per million persons and per GDP were calculated. Similar approaches were also used in other bibliometric analyses [15–18]. The number of publications for different organ, tissue, and cell transplant types was also quantified, using the adequate research strategy.

2.4. Statistical analysis

Data were analyzed using SPSS (Statistical Package for Social Sciences) version 22, where linear regression was used to relate the number of transplant publications to the average population and the average GDP.

2.5. Keyword and author co-occurrence analysis

Using the extracted data from the MEDLINE database, the VOS Viewer program (Version 1.6.15) analyzed the co-occurrence of keywords and the co-authorships of transplant-related publications in the Arab world. With a minimum threshold of 70 co-occurrent keywords, a bibliographic map was generated using the program's full counting method to visualize the most abundant keywords. Items of the map are gathered into non-overlapping clusters and linked by lines, which are assigned with values. The higher the value, the stronger the link of association/co-occurrence between items. Two types of visualization were used on VOS Viewer, the first being the network visualization, which attributes keywords into clusters, and the second being the overlay visualization, which assigns the average year of co-occurrence of each keyword. Besides, co-authorship analysis on VOS Viewer revealed the most prominent authors and institutions that contributed to transplant-related publications, as well as their link strength, which indicates the extent of collaboration between items.

3. Results

Table 1 shows the quantitative distribution of transplant-related publications for each Arab country between 2005 and 2019. 3118 articles that addressed transplantation were published by Arab countries, accounting for 1.56% of the total number of articles published by the Arab world (198,869) and for 1.25% of worldwide publications related to transplant during these 15-years.

During the timeframe of this study, 5 of all 22 Arab countries had no articles published concerning transplantation, including Comoros, Djibouti, Mauritania, Palestine, and Sudan. Egypt, contributing with 1002 of the totals, has the highest number of articles published in this field. This number accounts for nearly a third of the total Arab world transplant-related publications (32.13%). Saudi Arabia ranks second after Egypt with 922 transplant-related articles. Lebanon has the highest percentage of transplant-related articles out of its total publications (1.89%), followed by Egypt (1.88%).

Figs. 1 and 2 show the average population size and GDP in billion $ for each of the 22 Arab countries during the 15-year interval. The number of publications was normalized to the average population size and GDP to avoid bias when comparing results.

When normalized according to the average population size, Kuwait and Lebanon had the highest number of publications per million persons, 39.02 and 37.55, respectively. Besides the countries that had no publications, Yemen and Algeria had the lowest numbers with 0.23 and 0.76 publications per million persons, respectively.

Correlation between average population and number of transplant-

Table 1
Number of transplant-related articles in the Arab world.

Country	Number of publications on Transplantation	Number of total publications	% Transplant of Total
Algeria	31	4062	0.763
Bahrain	18	1460	1.233
Comoros	0	39	0.000
Djibouti	0	97	0.000
Egypt	1002	53,290	1.880
Iraq	57	4407	1.293
Jordan	139	10,817	1.285
Kuwait	140	5646	2.480
Lebanon	231	12,227	1.889
Libya	11	997	1.103
Mauritania	0	108	0.000
Morocco	55	7618	0.722
Oman	63	4835	1.303
Palestine	0	1042	0.000
Qatar	81	8265	0.980
Saudi Arabia	922	53,898	1.711
Somalia	0	94	0.000
Sudan	34	3069	1.108
Syria	25	1356	1.844
Tunisia	209	14,633	1.428
United Arab Emirates or UAE	94	9731	0.966
Yemen	6	1178	0.509
Total	3118	198,869	1.568
Worldwide	248,319	13,995,404	1.774

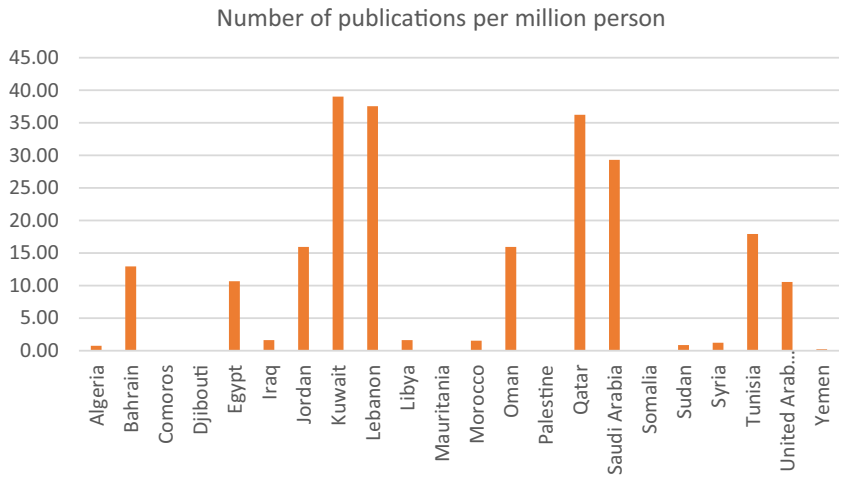


Fig. 1. Number of transplant-related articles per million persons.

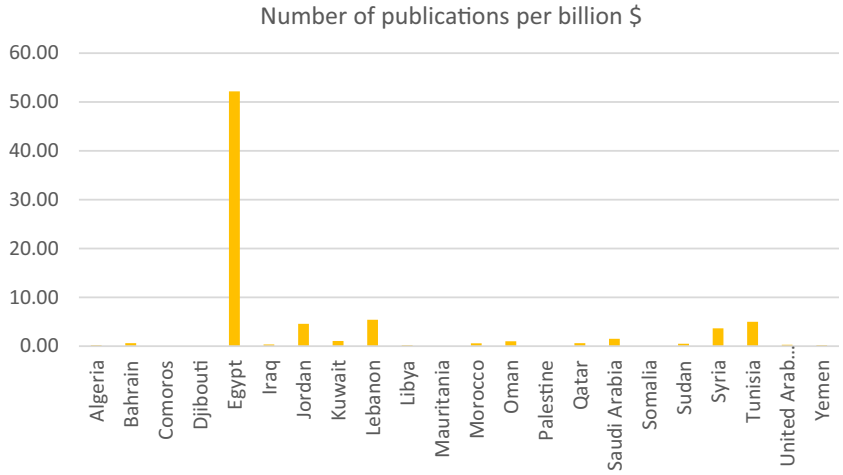


Fig. 2. Number of transplant-related articles per GDP.

related publications revealed a weak correlation ($R = 0.386$). R squared was equal to 0.15, indicating that the average population explains only 15% of the variation in transplant-related publications.

When normalized according to the average GDP, Egypt had the highest number of publications per billion dollars (52.17), with a noticeable difference with respect to the second, Lebanon (5.42). Algeria had the lowest number of publications per billion dollars (0.18), disregarding the countries with zero publications.

When looking for a correlation between GDP and the number of transplant-related publications, the *p*-value was less than 0.001, suggesting a significant relation. $R = 0.727$, showing a strong correlation. R squared = 0.592, showing that GDP can explain 59.2% of the number of

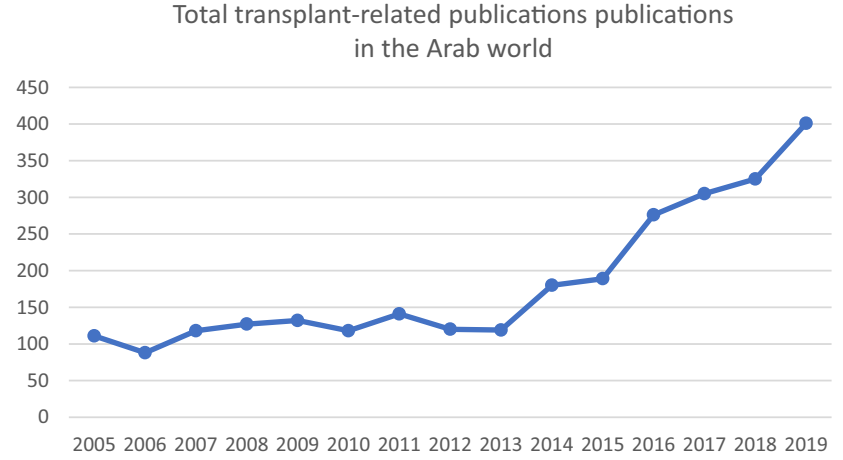


Fig. 3. Line graph showing the number of total transplant-related publications in the Arab world per year.

transplant-related publications.

The number of transplant publications had a growing inflection in 2013, with the last six years alone witnessing 65.27% of the total Arab transplant-related publications Fig. 3.

Transplant-related publications stratified by organ, tissue, and cell transplant types were shown in Table 2. Regarding organs, the kidney had the highest number of related articles with 129. Concerning tissues, the bone had the highest number of publications (199), followed by the vessel (105) and the cornea (73). Stem cells had 209 publications, the highest cell-related number of publications.

Regarding the most productive Arab authors in the transplantation field, a maximum of 73 papers were published by the same author (Aljurf, Mahmoud), followed by Ghoneim Mohamed A with 57 publications. The maximum total link strength was observed in the third most prominent author, Gheith, Osama A (link strength = 134), followed by Halim, Medhat A with a 131 total link strength. In total, only 15 authors have more than 33 transplant-related publications with total link strength varying between 45 and 134, showing a wide variety of co-author collaborations Table 3.

Regarding the most productive Arab organizations in terms of transplant-related publications, only five institutions had a total of 15 or more transplant-related publications. A maximum of 51 papers was published by the same organization (King Faisal Specialist Hospital), followed by the Mansoura university urology and nephrology center in Egypt with 32 publications. Strength links varied between 0 in Mansoura University, Egypt, indicating no collaborations with other institutions and 6 in King Faisal research center. Further analysis on VOS Viewer revealed a modest cluster of co-authorship centered by King Faisal University and associated with the King Saud College of Medicine, the American University of Beirut, and the Bone Marrow and Stem Cell Transplant Unit in Jordan. These results indicate poor institutional collaborations and only modest cross-country collaborations between Jordan, Saudi Arabia, and Lebanon Table 4.

Network visualization of MeSH keywords co-occurrence revealed four main clusters: the first one, in red color, focused on tissue-related keywords and contained the following keywords: tissue donors, tissue and organ procurement, graft survival, and biopsy. The second cluster, indicated by the blue color, focused on bone marrow transplant, as it contained HLA antigens, histocompatibility testing, transplantation homologous, and bone marrow transplant. The third cluster, in green color, encompassed the epidemiological aspect of transplant and included age, prospective, retrospective, male, female, and follow-up studies keywords. The last cluster, colored in yellow, focused on mesenchymal stem cells and included stem cell transplantation, mesenchymal stem cell, and biomarkers keywords. (Fig. 4).

The co-occurrence of MeSH keywords using the overlay visualization option on VOS Viewer revealed that publications in the Arab world shifted from focusing on solid organ transplant in 2013, particularly kidney and skin transplant, towards histocompatibility analysis, survival, and epidemiology in 2014, followed by experimental studies (mice and animals keywords) and hematopoietic stem cell transplant in 2015, and mesenchymal stem cells in 2016. These results show a modest but increasing trend towards experimental work on animals and newer therapies such as mesenchymal stem cells in the field of transplant in the Arab world Fig. 5.

Table 3
Top 15 most productive authors in the field of transplant in the Arab world.

Author's Name	Number of publications	Total link strength
Aljurf, Mahmoud	73	63
Ghoneim, Mohamed A	57	84
Gheith, Osama A	49	134
Halim, Medhat A	43	131
Abderrahim, Ezzedine	41	50
Al-Otaibi, Torki	40	128
Ayas, Mouhab	40	75
Bazarbachi, Ali	39	28
Nair, P	38	128
Said, Tarek	38	122
Bakr, Mohamed Adel	36	45
Al-Seraihy, A	34	65
Ben Abdallah, Taieb	34	46
Nampoory, Narayanan	33	88
Mohty, Mohamad	32	54

Table 4
Top 5 most productive organizations/institutions in the field of transplant in the Arab world.

Organization's Name	Number of papers	Total link strength
King Faisal specialist hospital and research center, Riyadh, Saudi Arabia	51	6
Urology and nephrology center, Mansoura University, Mansoura, Egypt	32	0
Mansoura university, Mansoura, Egypt	26	6
Bone marrow transplantation program, department of internal medicine, American University of Beirut Medical Center, Beirut, Lebanon	18	2
College of Medicine, Alfaisal University, Riyadh, Saudi Arabia	15	4

4. Discussion

Despite having imperative live-saving and lifestyle-improving roles, transplantation is still far from being an essential topic in the Arab world. This hypothesis is supported by the low numbers of published transplant-related articles over the past 15 years, neither exceeding 2% of the total number of articles published by the Arab world nor the worldwide transplant-related publications.

Nevertheless, since 2013, the number of transplant publications in the Arab world showed a growing inflection, despite the numerous countries that did not participate in this growth, such as Comoros,

Table 2
Number of total publications per transplant type in the Arab world.

Organs	Number of publications	Tissues	Number of publications	Cells	Number of publications
Heart	38	Cornea	73	Bone marrow	18
Lung	13	Skin	34	Stem cell	209
Liver	74	Heart valve	16	Umbilical cord stem cells	29
Pancreas	22	Nerves	0	Blood stem cells	82
Uterus	8	Vessel	105		
Kidney	129	Bone	199		
Trachea	2	Tendon	23		
Thymus	0	Islets of Langerhans	18		
Intestine	21	Limbs	33		
Stomach	2	Face	19		
Testis	17				
Penis	22				

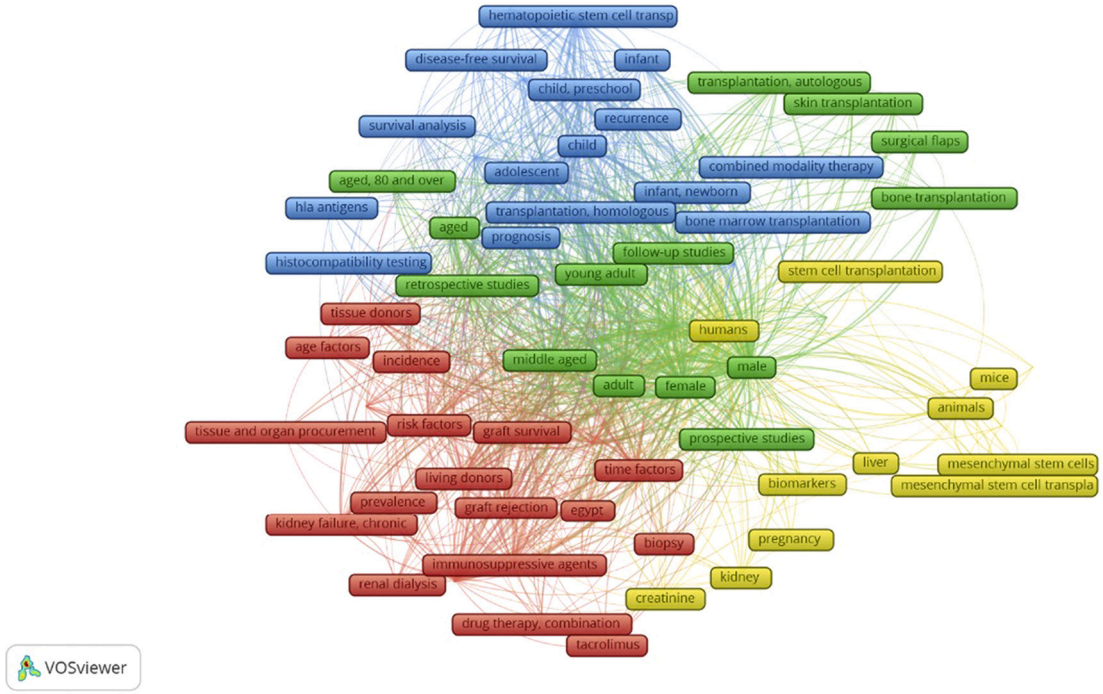


Fig. 4. Co-occurrence of MeSH keywords related to transplant from 2005 till 2019 using VOS Viewer network visualization option.

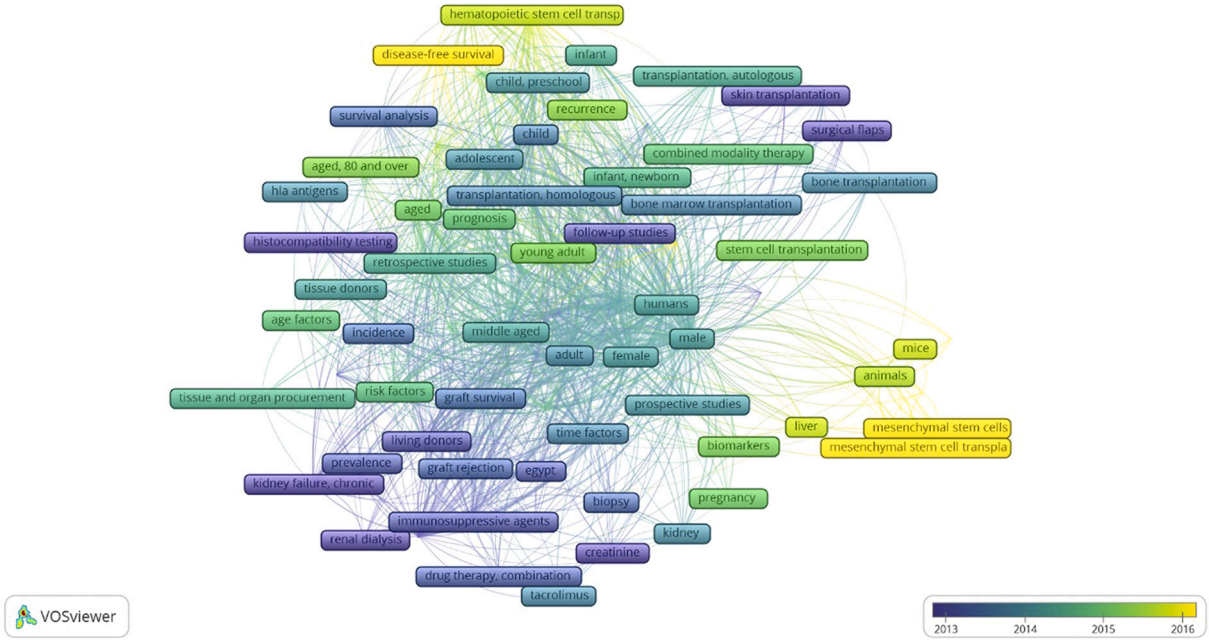


Fig. 5. Co-occurrence of MeSH keywords related to transplant using VOS Viewer overlay visualization option.

Djibouti, Mauritania, Palestine & Somalia. These countries had absolutely no publications in this domain, which was not surprising due to their low GDP, a factor associated with a negative impact on the number of published papers [19,20].

Two metrics were used in the comparison of research activity in the transplantation field between countries. Kuwait and Lebanon showed the highest publications number per average population, with 39.02 and 37.55 publications per million persons, respectively. In opposition to psoriasis and Familial Mediterranean Fever publications [16,18], the average population size showed a low correlation with the number of transplant publications. Egypt had the highest number of publications according to the average GDP as it had 52.17 publications per billion dollars. Lebanon appeared second with a tremendous difference from Egypt since it had only 5.42 publications per billion dollars. Similar to stroke and atrial fibrillation publications [15,17], the number of transplant publications strongly correlated with GDP, making this metric more reliable and accurate in assessing transplant-related publications.

There is a noticeable fluctuation in transplant research between Arab countries due to the varying economic, social, medical, and legal situations [21]. Despite this fact, which shows the necessity of collaboration between authors and organizations, collaboration is minimal in Arab countries, as depicted by this study's results.

Mainly, transplant research in the Arab world focused on tissues, bone marrow, epidemiology, and mesenchymal stem cells. These topics have gained focus recently, as research orientation towards experimental studies has increased. Solid organ transplant and histocompatibility were studied in an earlier era.

In general, Muslim communities tend to have a negative opinion towards organ transplantation, showing that this is a crucial ethical dilemma [22,23]. In Islam, there is a debate about the concept and definition of brain death [24]. The fact that neither *Qur'an* nor *Sunnah* directly addresses brain death had led to the absence of scholarly consensus [25], making the transplantation-related rulings case-specific and not in a global context [26]. Thus the negative attitude of Muslims regarding transplantation is attributed to various aspects and not a unique religious issue, like mistrust of the healthcare system, lack of information regarding organ donation, and lack of clear understanding of religious rulings. Partnering with religious leaders to address transplant-related affairs may help reach more positive attitudes, as suggested by some studies [27].

Nevertheless, the religious barrier is certainly not the sole reason for attitudes' variability. It is worth mentioning that Saudi Arabia has one of the highest rates of organ transplant activity [11] and has accounted for most of the deceased donor thoracic and abdominal organ transplantations in the region [28].

Additionally, we can also talk about the restraints of hematopoietic stem cells transplantation procedures mainly due to financial states [29], as the religious point of view is clear regarding the acceptance of these practices [30]. This issue has a pronounced effect on regional morbidity and mortality because of the widespread inherited genetic diseases and the contribution of stem cell transplant to their treatment [31,32]. We mention the immense progress Saudi Arabia had in this field [33].

Some sociologists linked the unwillingness to participate in organ transplant procedures to the lack of organ donation culture in Arab societies, compared to Western countries who devoted tremendous efforts to promoting transplantation [34]. For instance, the deficiency in understanding and accepting the concept of brain death in Arab countries had a consequential impact on the number of potential donors [35].

When taking the example of Egypt, we can correlate the high publication count with the social and national necessity. Egypt is the most affected country by hepatitis C [36], which raises the need for liver transplantation in patients suffering end-stage liver diseases [37], despite the limited possibilities for these procedures [38].

Several risk factors in the Arab world predispose to eventual organ transplantation. Diabetes mellitus and hypertension lead to end-stage renal diseases [39], hepatitis B & C infections leading to end-stage liver diseases [40], urbanization, and sedentary lifestyle leading to end-stage heart failure [41]. These end-stage diseases require creating an appropriate environment for organ transplantation procedures, which is a potentially practical and definitive treatment option for patients with severe organ dysfunctions [42].

In summary, the progress in the transplantation field in the Arab world is facing numerous barriers. Despite the past few years showing an improved transplant-related research activity in both quantity and quality, many issues should yet be addressed to achieve a suitable and comfortable environment for transplantation procedures. Socioeconomic and healthcare system amelioration and improved role of religious leaders in addressing transplant concerns are indispensable to achieve this goal.

To the best of the authors' knowledge, this is the first study evaluating the regional contribution of Arab countries to the field of organ, tissue, and cell transplantations and the publication distribution concerning different types of procedures. However, some limitations should be noted. A unique database (PubMed) was used to obtain the publications. Furthermore, only papers written in the English language were gathered, excluding publications written in Arabic or other languages, leading to a possible effect on the reliability of the results as the number of publications is undervalued. Nevertheless, since most papers are published in the English language and cited through PubMed, these limitations are minimal, and the comparison of Arab countries' results rigorously describes their research activity.

5. Conclusion

Despite its growth in recent years, transplant research in the Arab world remains a modest contributor to the field of transplantation due to various economic, social, and religious factors affecting research activity. This paper collects background information to be used by healthcare personnel in order to improve and create better conditions for future transplantation procedures.

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Declaration of competing interest

We declare no conflict of interest.

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